

Homework 7

Instructions

Complete the following problems and submit to Gradescope your work as a file named `hw7.Rmd`.

A file with the wrong name or a bug will prevent the autograder from being able to run your submission. Check that your filename is correct, and check for bugs clearing your memory and then running your file from start to finish. It's also a good idea to check that your file "knits."

Note: Do not use any loops and hard-code as little as possible. For all of these problems use a significance level of $\alpha = .05$

Problem 1

Consider a chi-squared distribution with 2 degrees of freedom. Plot the density curve of the population distribution and describe its shape. Assign your plot object to a variable called `myPlot`

Problem 2

Consider a chi-squared distribution with 2 degrees of freedom, which has a population mean of 2 and a population standard deviation of 2. Use Monte Carlo simulation with 10,000 repetitions to determine the empirical Type I error of the correctly conducted z-test as described in each part for sample sizes 8, 23, and 52.

Z-tests are the correct hypothesis test if your data are normally distributed (which isn't true in this case), or if your data sample is sufficiently large.

a: the two-sided case

Write a function called `do.two.sided.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `z.two`. Have them be stored in the order mentioned above.

b: the left-sided case

Write a function called `do.left.sided.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `z.left`. Have them be stored in the order mentioned above.

c: the right-sided case

Write a function called `do.right.sided.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 vector called `z.right`. Have them be stored in the order mentioned above.

Problem 3

Create a labeled tabular structure to display the results from Problem 2. What conclusions can you draw?

Specifically, this tabular structure should be a `matrix` called `prob3Mat`. Its first row should correspond with the results obtained for the sample size of 8. The second row should correspond with the results obtained for the sample size of 23, etc. Make sure to change the column names of this object to `c("two", "left", "right")`.

```
alt <- matrix(c(z.two, z.left, z.right),
              nrow = length(z.two),
              ncol = 3,
              byrow = FALSE)

colnames(alt) <- c("two", "left", "right")

alt
```

```
##           two  left  right
## [1,] 0.0446 0.0223 0.0673
## [2,] 0.0463 0.0328 0.0609
## [3,] 0.0475 0.0404 0.0585
```

Problem 4

Consider a chi-squared distribution with 2 degrees of freedom, which has a population mean of 2 and a population standard deviation of 2. Use Monte Carlo simulation with 10,000 repetitions to determine the empirical Type I error of the correctly conducted t-test as described in each part for sample sizes 8, 23, and 52.

a: the two-sided case

Write a function called `do.two.sided.t.test()` (**note the difference in the name!**) that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the t-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 vector called `t.two`. Have them be stored in the order mentioned above.

b: the left-sided case

Write a function called `do.left.sided.t.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the t-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `t.left`. Have them be stored in the order mentioned above.

c: the right-sided case

Write a function called `do.right.sided.t.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the t-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `t.right`. Have them be stored in the order mentioned above.

Problem 5

Create a labeled tabular structure to display the results from Problem 4. What conclusions can you draw?

Specifically, this tabular structure should be a **matrix** called `prob5Mat`. Its first row should correspond with the results obtained for the sample size of 8. The second row should correspond with the results obtained for the sample size of 23, etc. Make sure to change the column names of this object to `c("two", "left", "right")`.

Problem 6

Consider a chi-squared distribution with 2 degrees of freedom, which has a population mean of 2 and a population standard deviation of 2. Use Monte Carlo simulation with 10,000 repetitions to determine the empirical Type I error of the incorrectly conducted z-test (sample standard deviation used) as described in each part for sample sizes 8, 23, and 52.

a: the two-sided case

Write a function called `do.two.sided.bad.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `zt.two`. Have them be stored in the order mentioned above.

b: the left-sided case

Write a function called `do.left.sided.bad.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 **vector** called `zt.left`. Have them be stored in the order mentioned above.

c: the right-sided case

Write a function called `do.right.sided.bad.test()` that takes as an argument the sample size of your data, and returns either `TRUE` or `FALSE` depending on whether or not the z-test mentioned above rejects the null or fails to reject it.

Collect the three empirical proportions in a length 3 vector called `zt.right`. Have them be stored in the order mentioned above.

Problem 7

Create a labeled tabular structure to display the results from Problem 6. What conclusions can you draw?

Specifically, this tabular structure should be a `matrix` called `prob7Mat`. Its first row should correspond with the results obtained for the sample size of 8. The second row should correspond with the results obtained for the sample size of 23, etc. Make sure to change the column names of this object to `c("two", "left", "right")`.